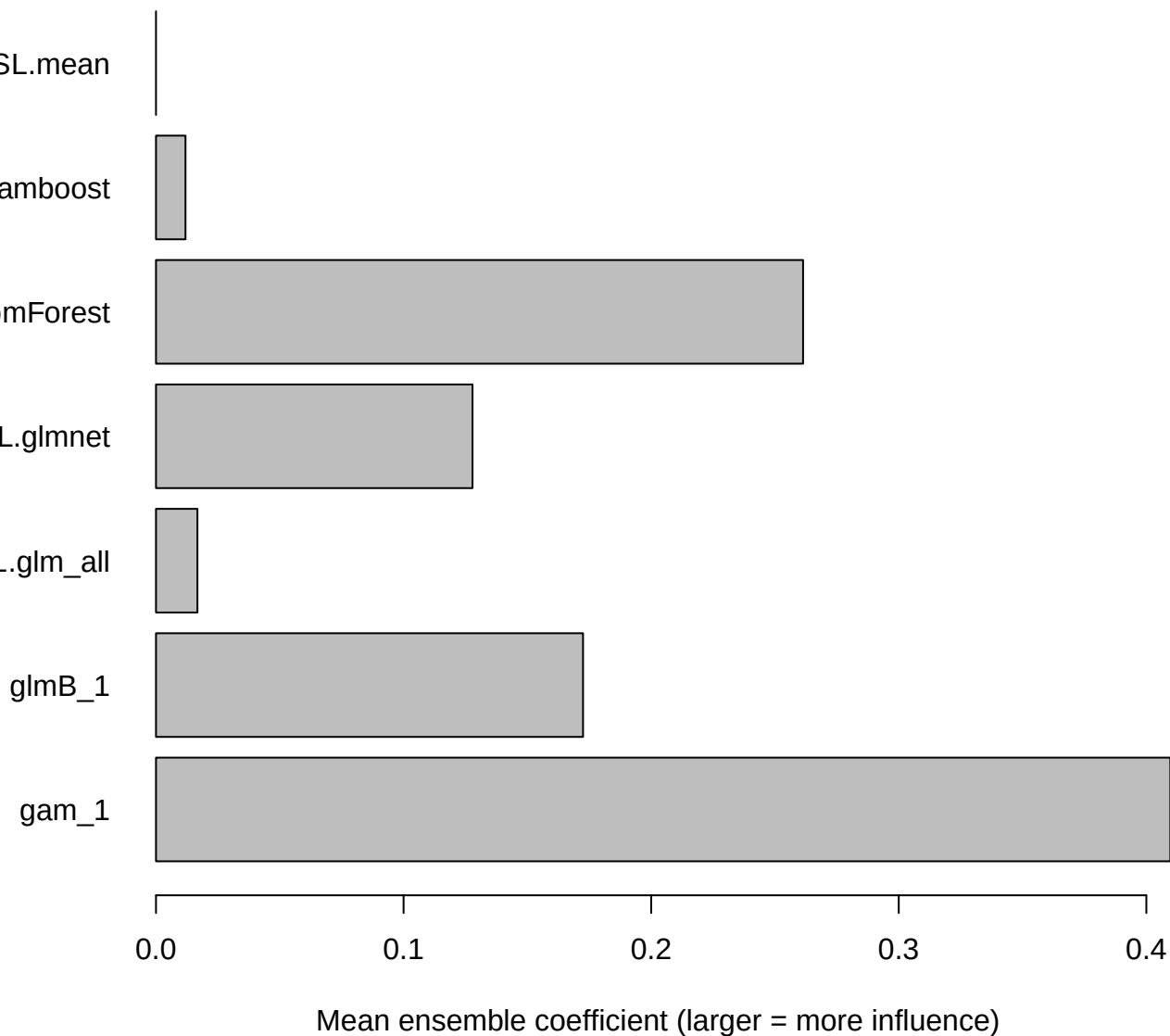
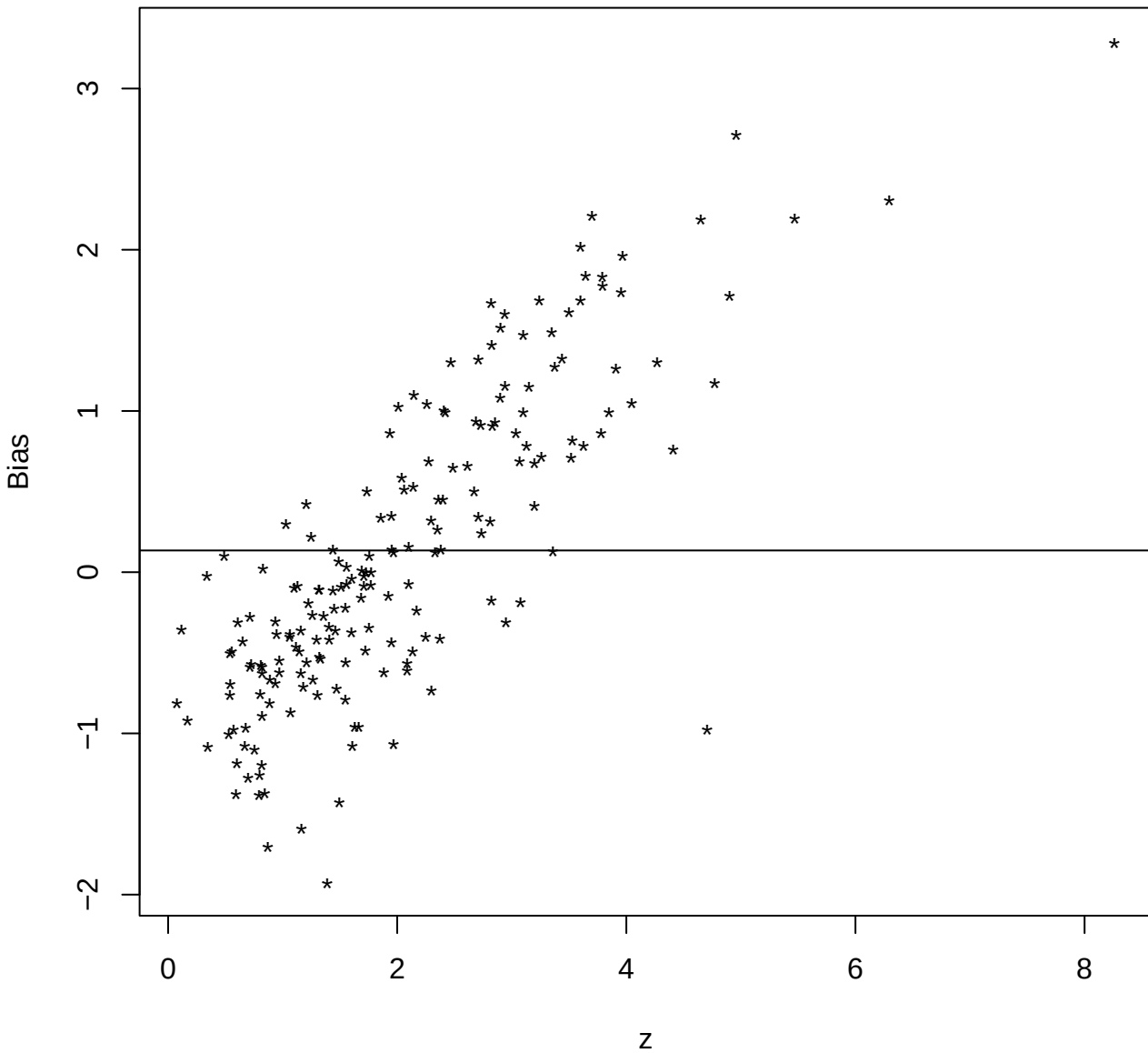


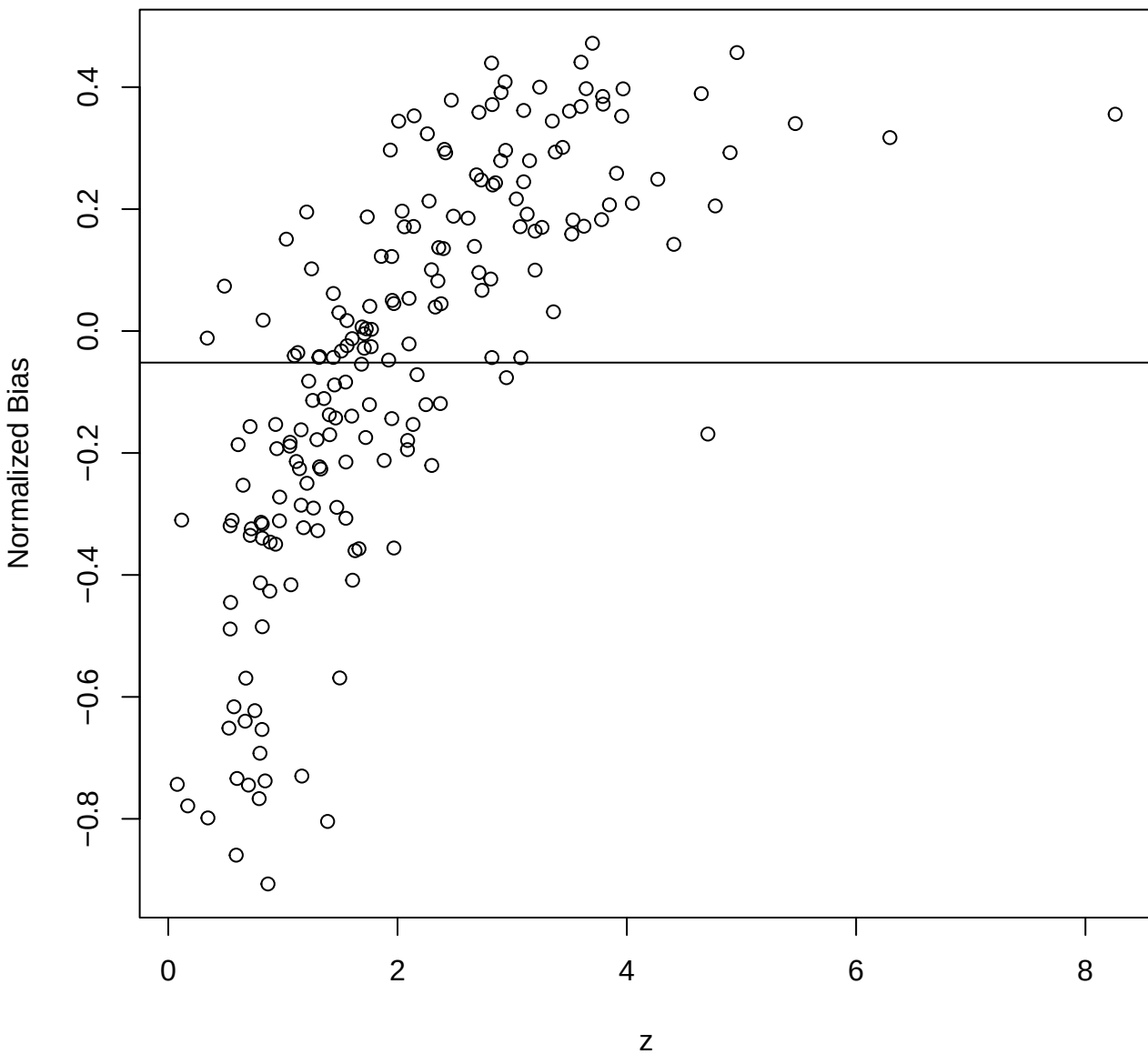
### SuperLearner ensemble weights (unsorted)

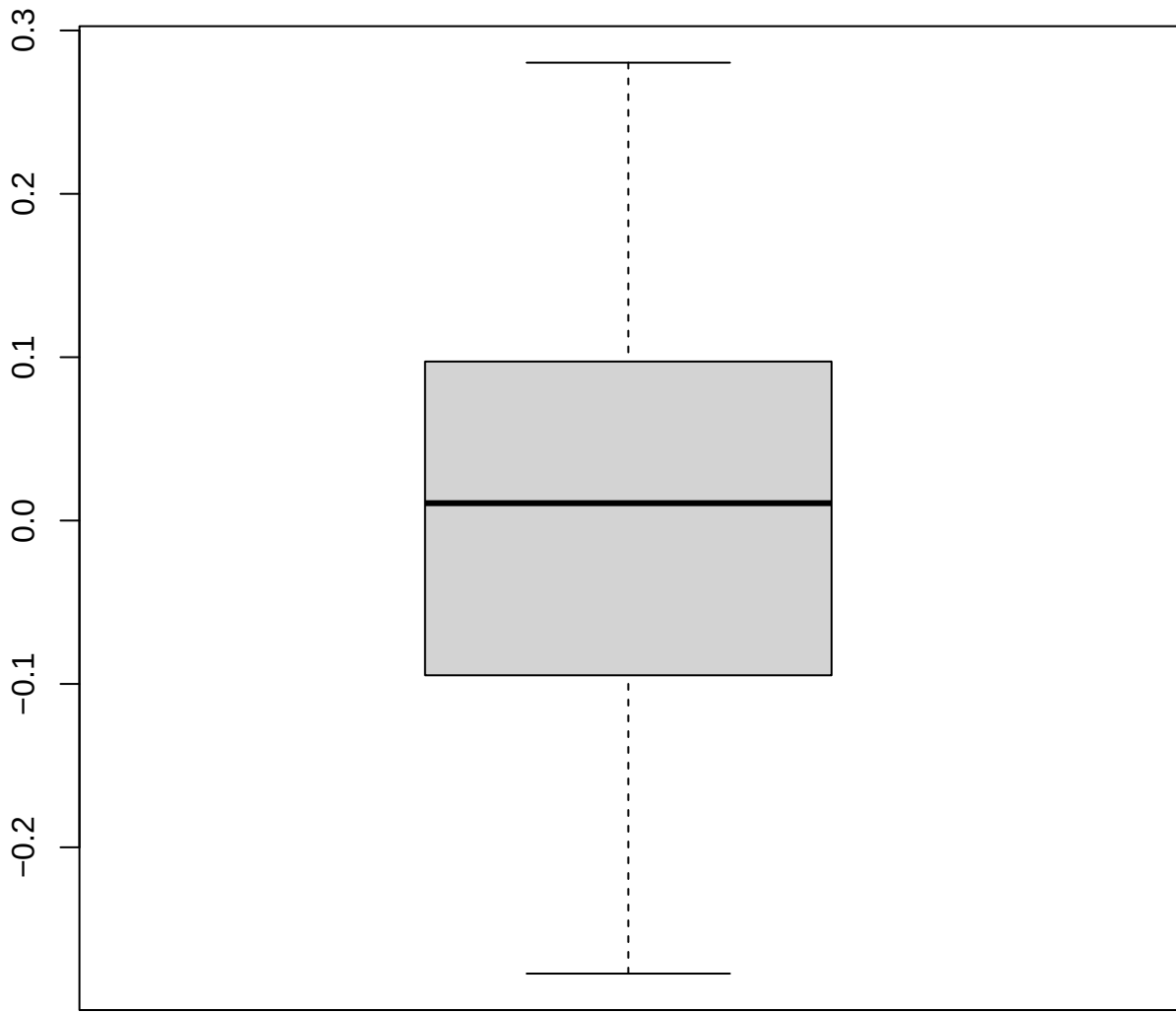


# Redshift vs Bias



**Redshift vs Normalized Bias**

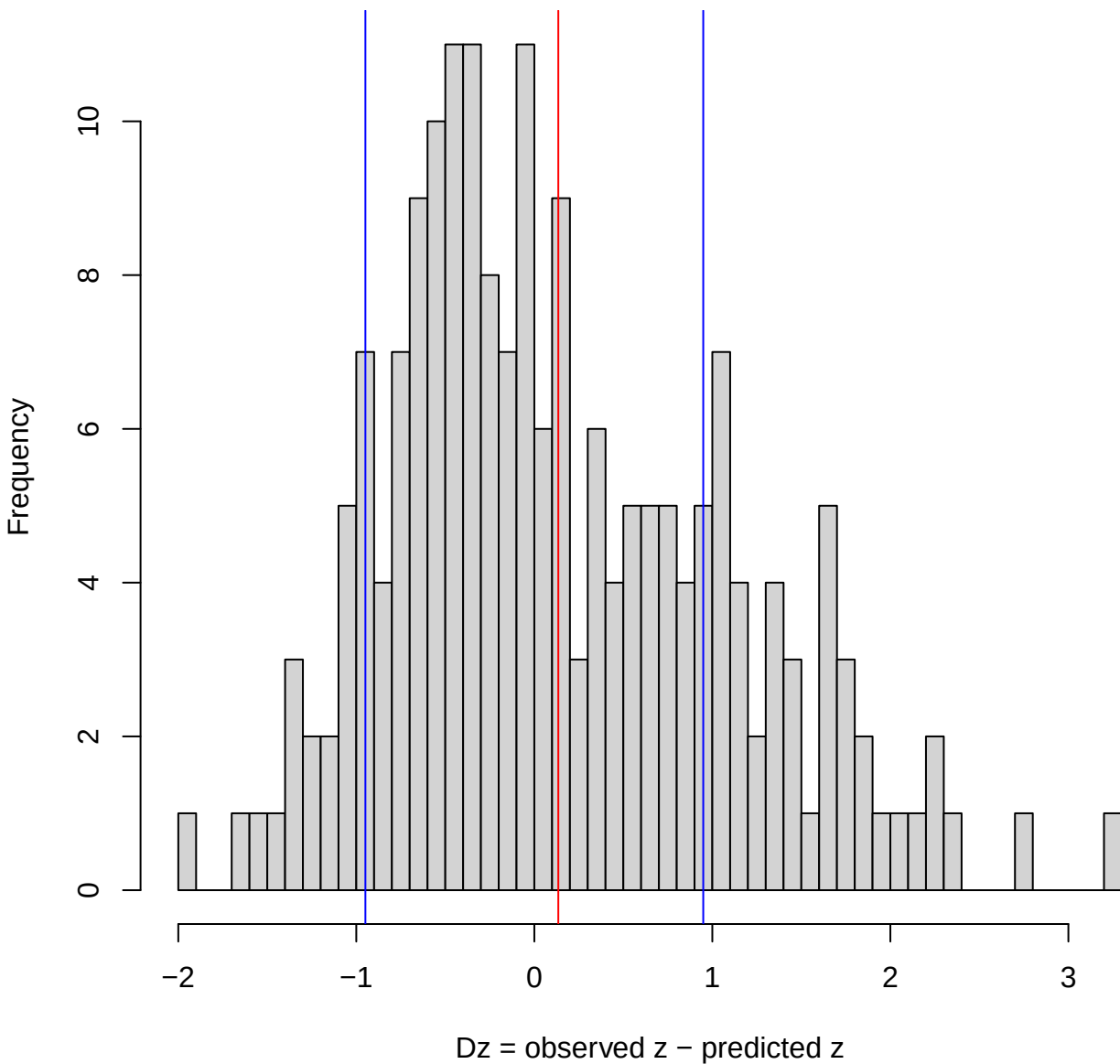




**Redshift residual distribution ( $Dz = \text{obs } z - \text{pred } z$ )**

**Sigma=0.949, Bias=0.135**

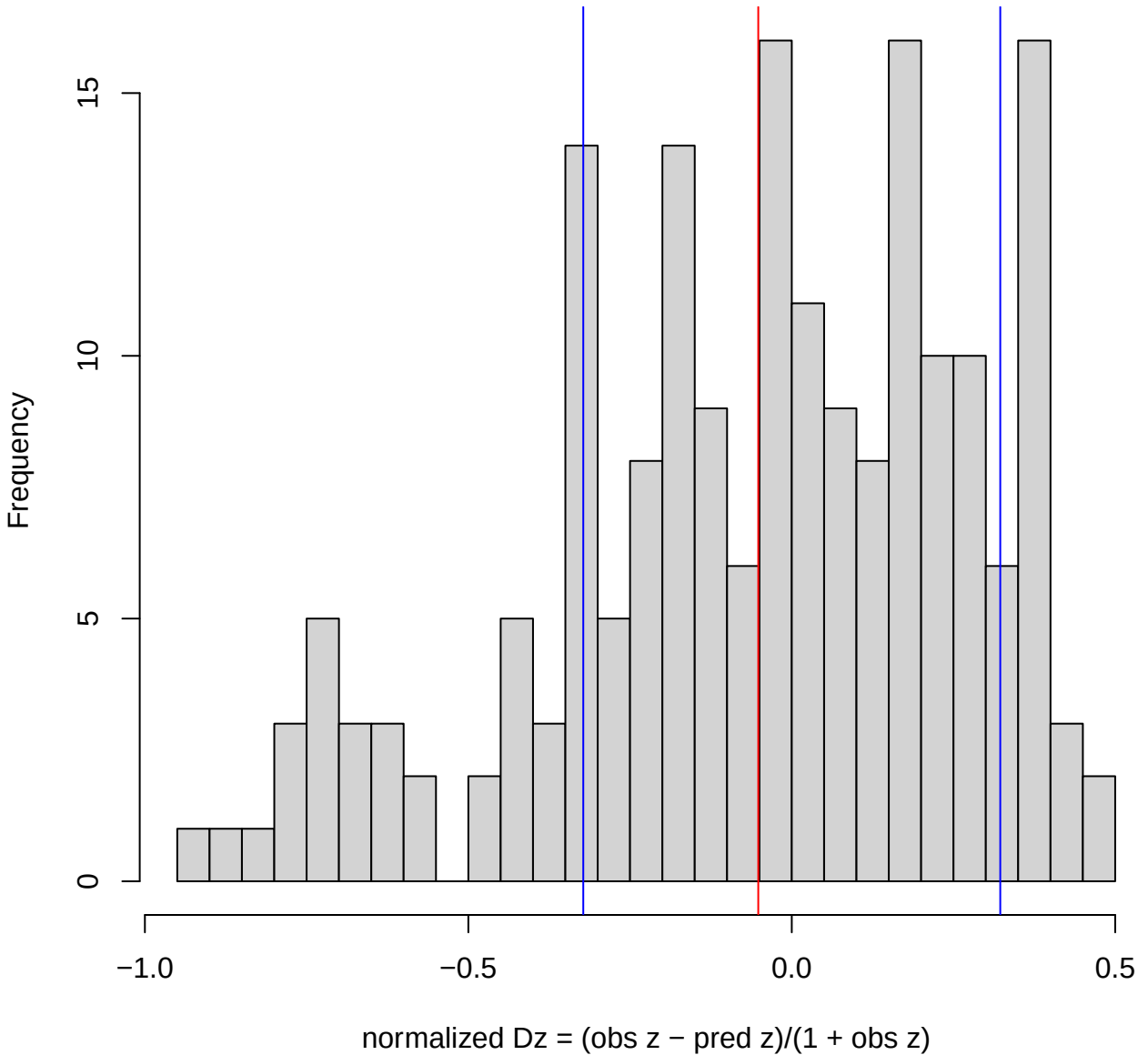
**(blue =  $\pm$ -Sigma, red = Bias)**



**Normalized residual distribution (normalized  $D_z = (\text{obs } z - \text{pred } z)/(1 + \text{obs } z)$ )**

**Sigma=0.322, Bias=-0.0518**

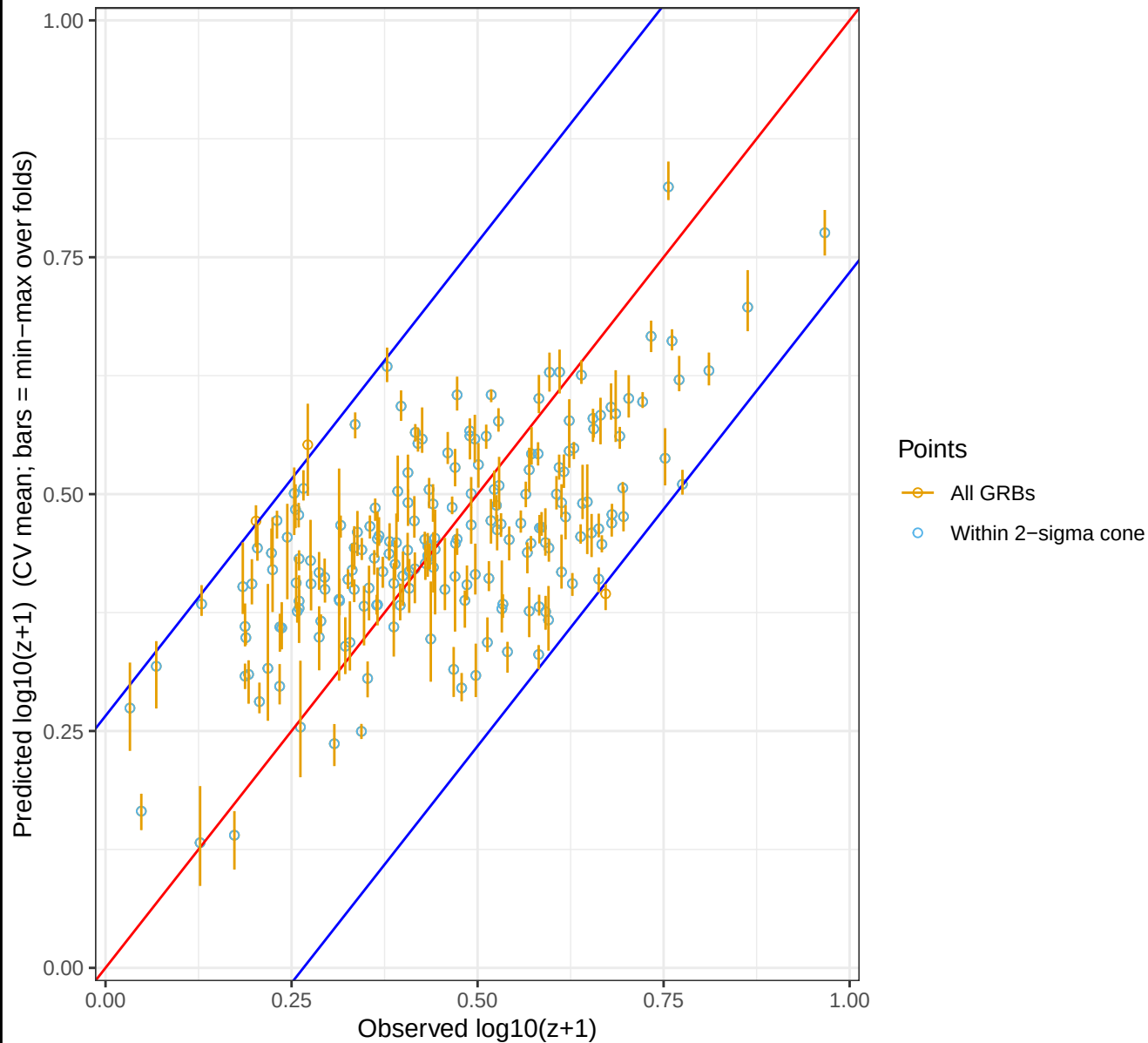
**(blue =  $\pm$ -Sigma, red = Bias)**



Cross-validated predicted vs. observed  $\log_{10}(z+1)$

Sample = 192 GRBs | within 2-sigma cone = 189 (98%)

$r = 0.6288$  |  $\text{Sigma} = 0.133$  |  $\text{RMS} = 0.133$  |  $\text{Bias} = -0.0018$  |  $\text{NMAD} = 0.142$

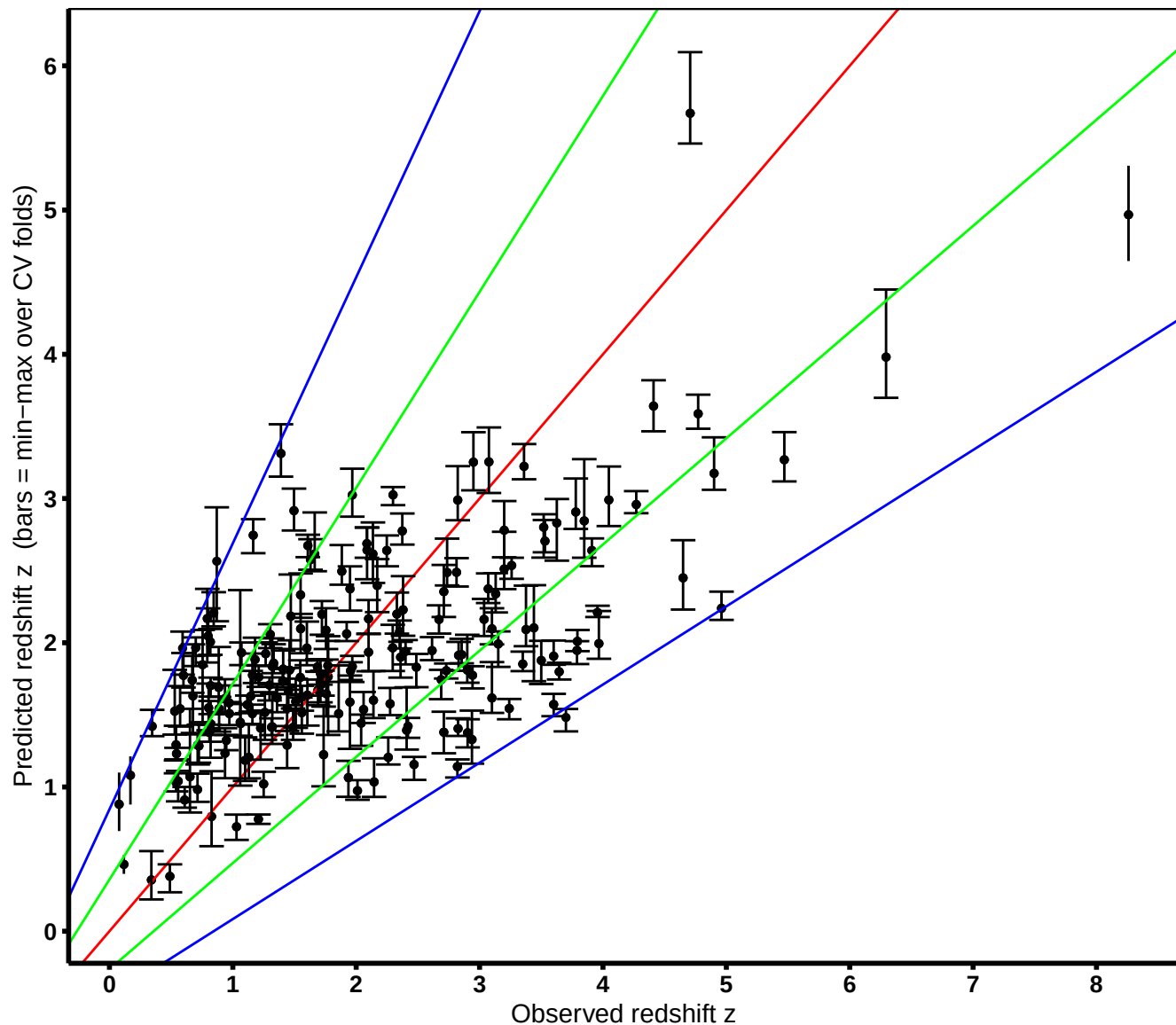


Cross-validated predicted vs. observed redshift  $z$

Sample = 192 GRBs | within 2-sigma cone = 189 (98%)

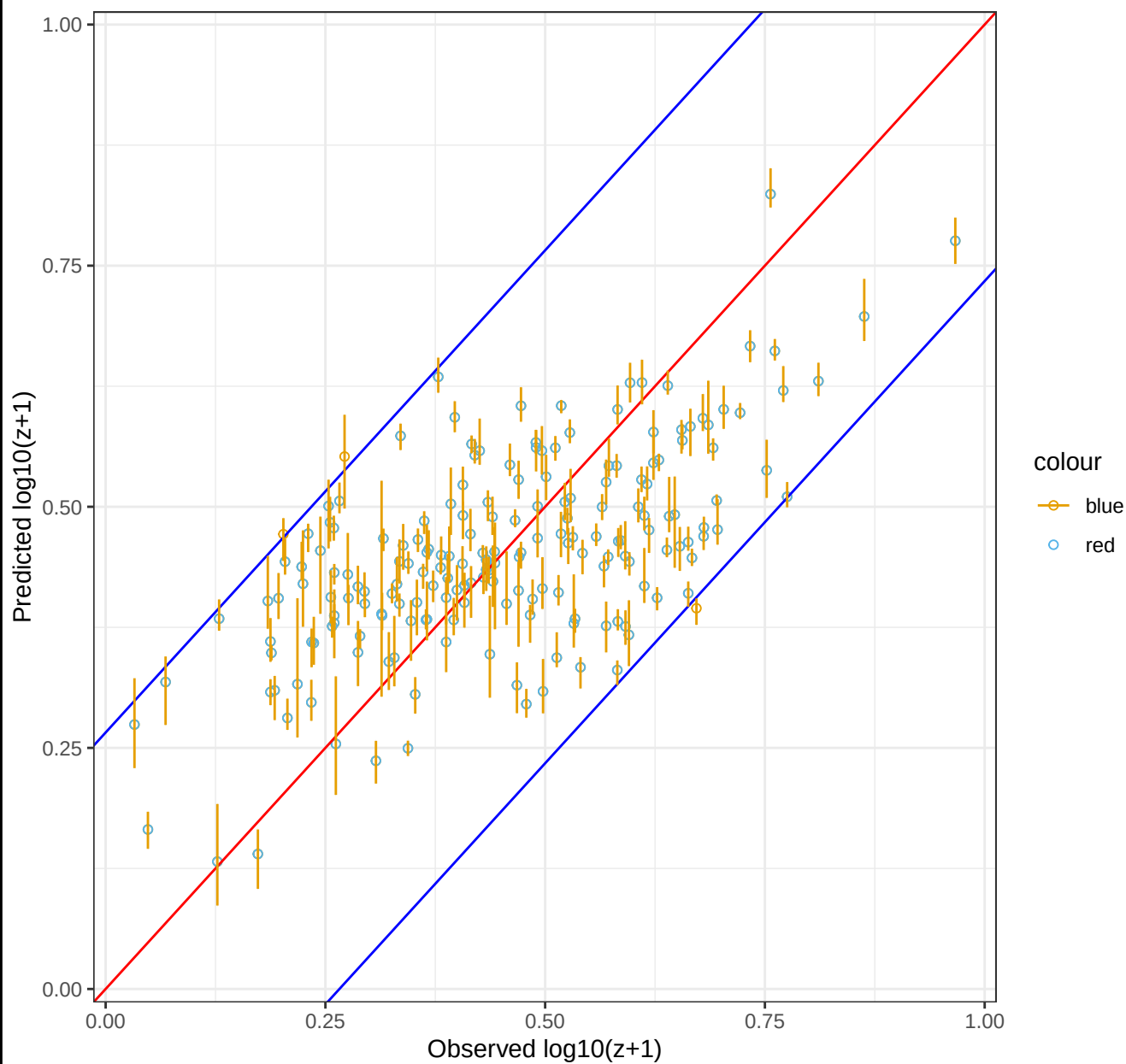
within 1-sigma cone = 130 (68%)

$r = 0.655$  |  $\text{Sigma} = 0.949$  |  $\text{RMS} = 0.96$  |  $\text{Bias} = 0.13$  |  $\text{NMAD} = 1.31$

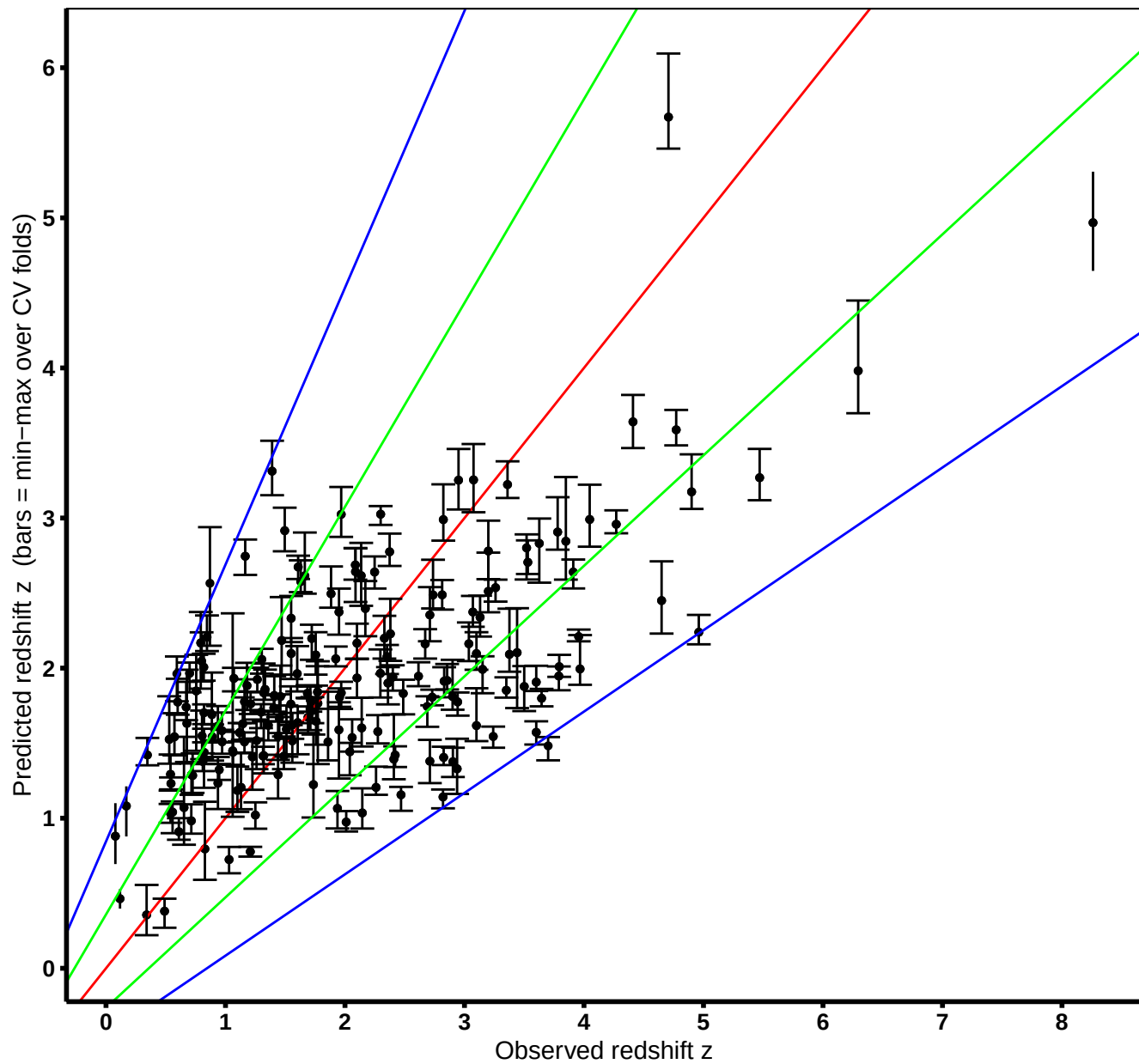




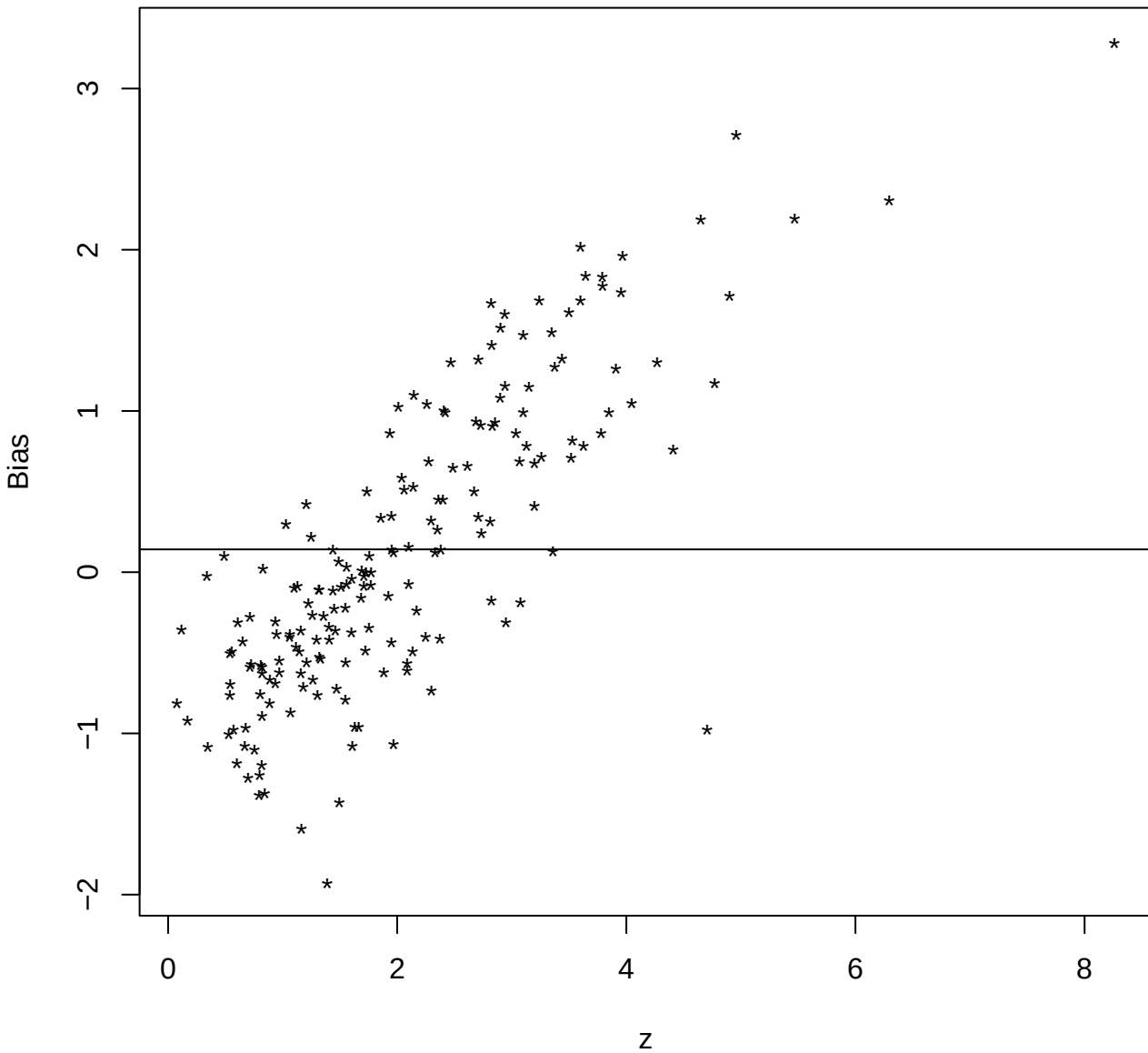
Predicted vs. observed  $\log_{10}(z+1)$   
(cross-validated)



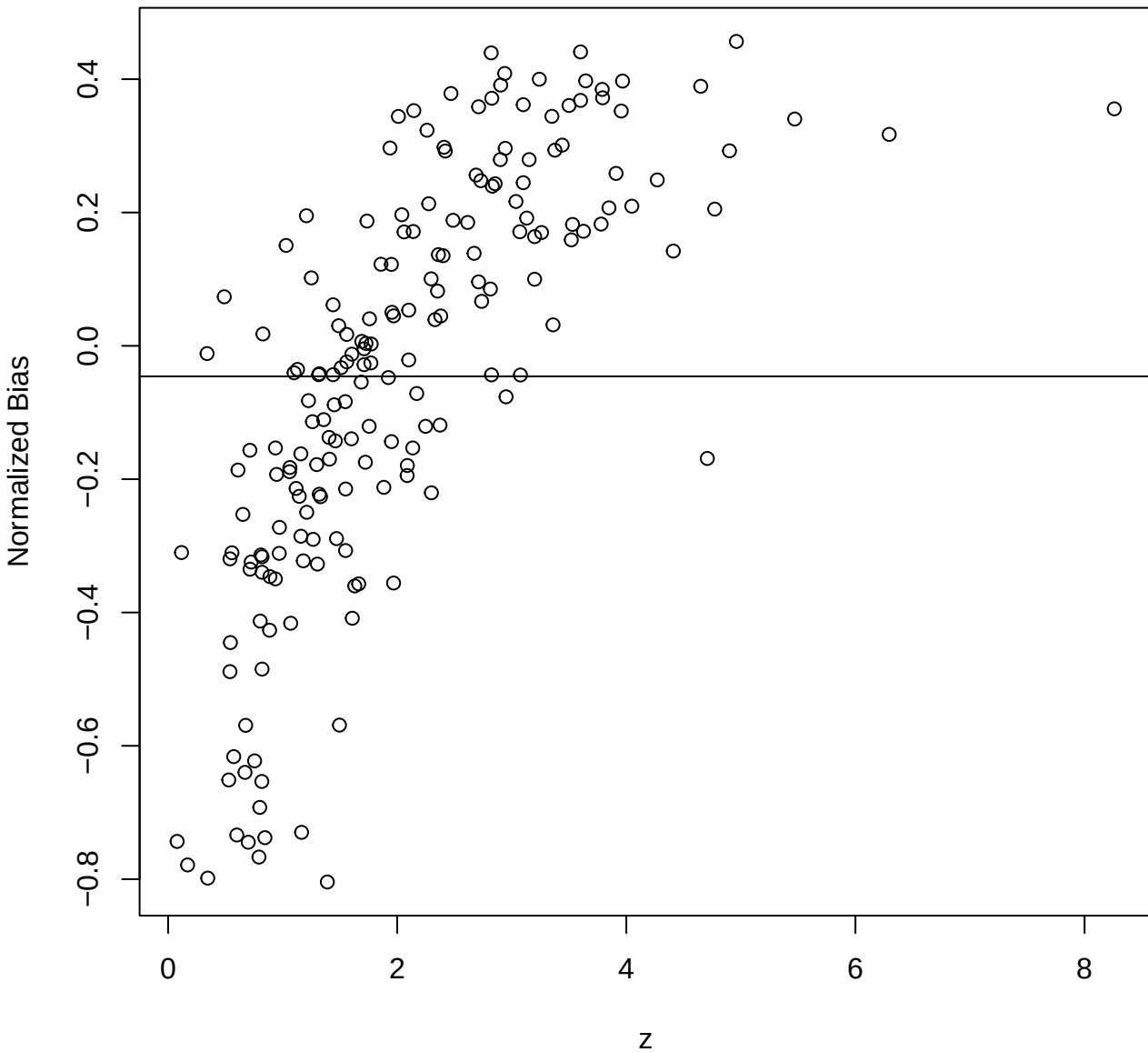
Predicted vs. observed redshift  $z$   
(cross-validated)

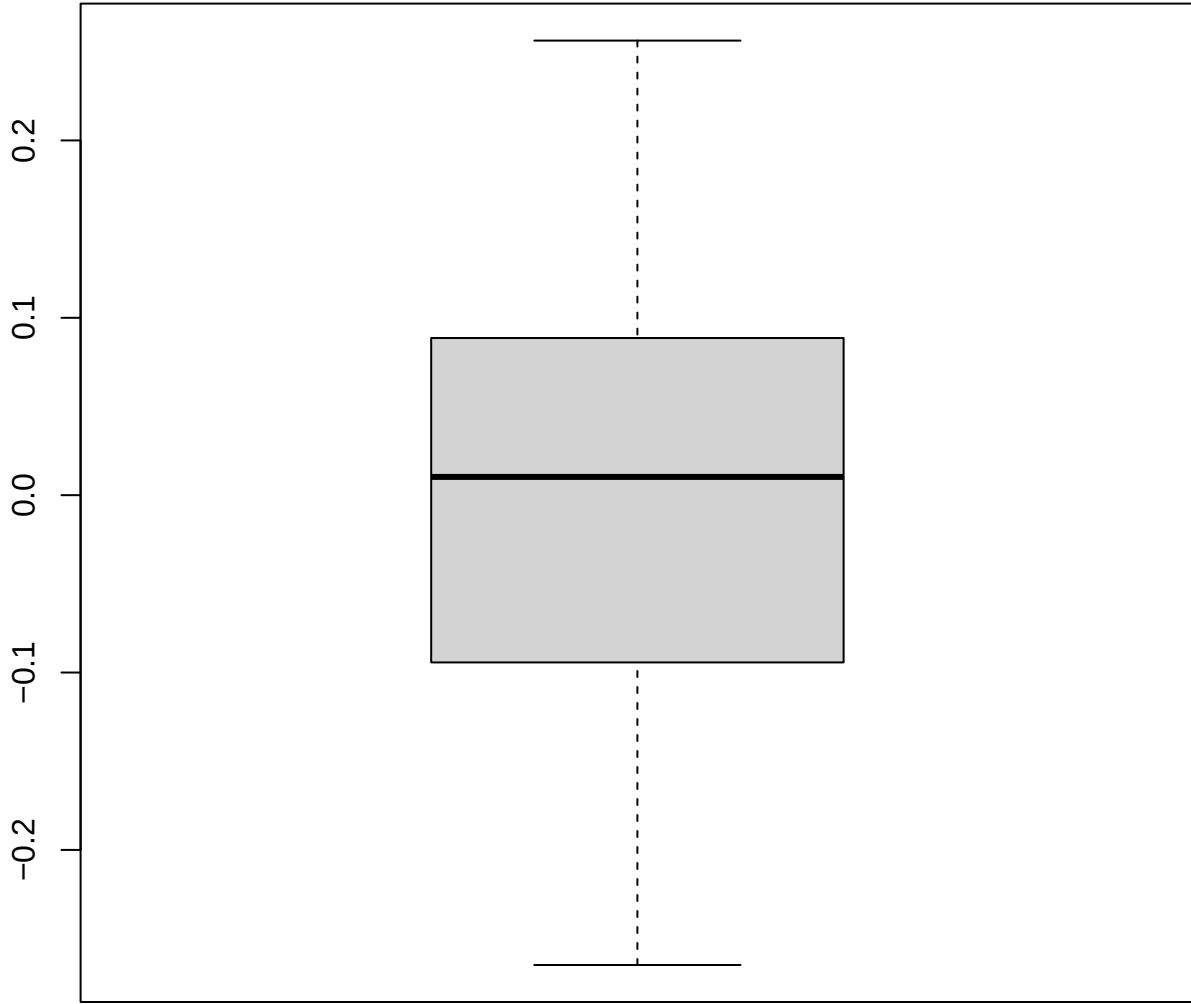


# Redshift vs Bias



**Redshift vs Normalized Bias**

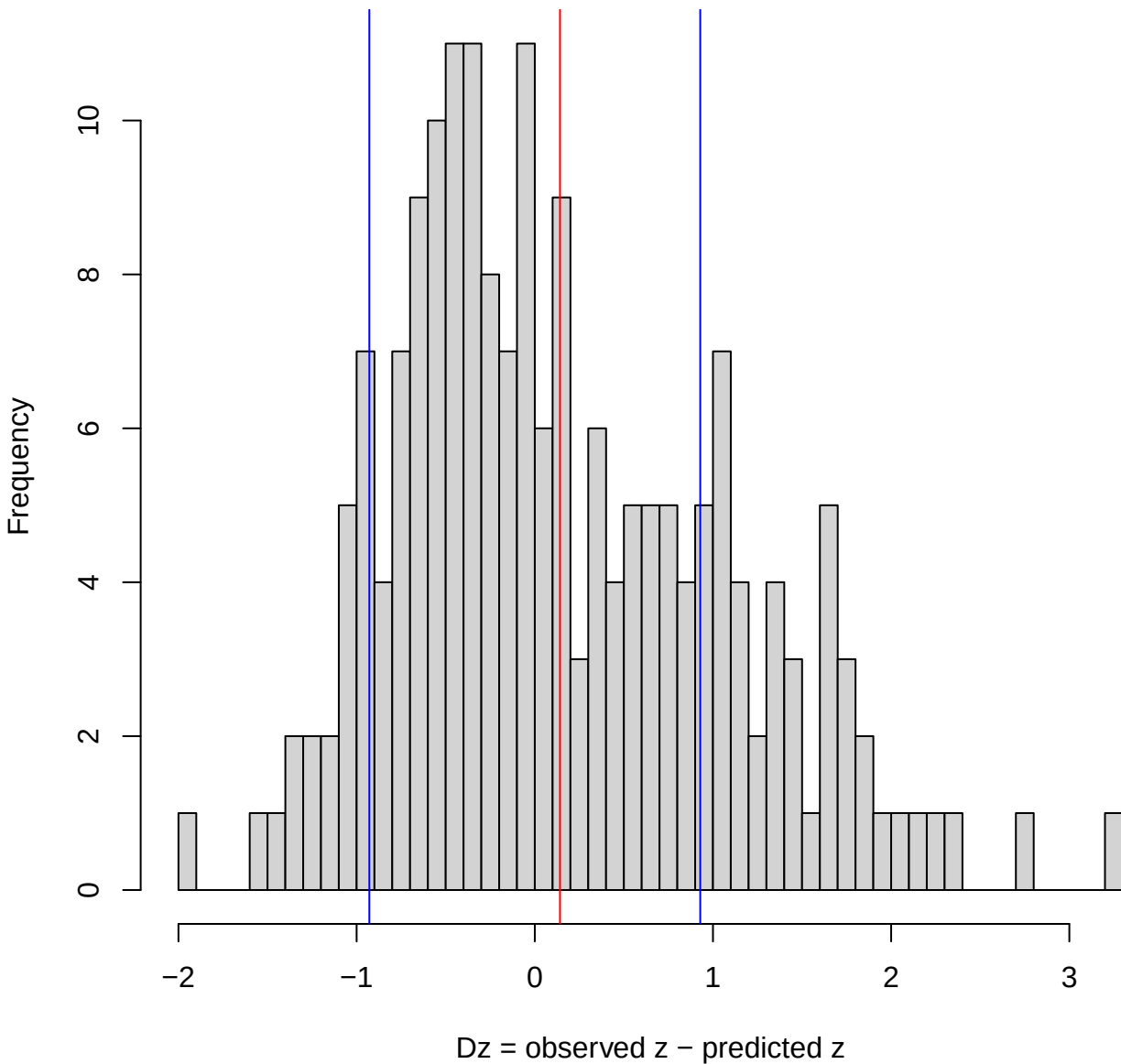




**Redshift residual distribution ( $Dz = \text{obs } z - \text{pred } z$ )**

**Sigma=0.928, Bias=0.141**

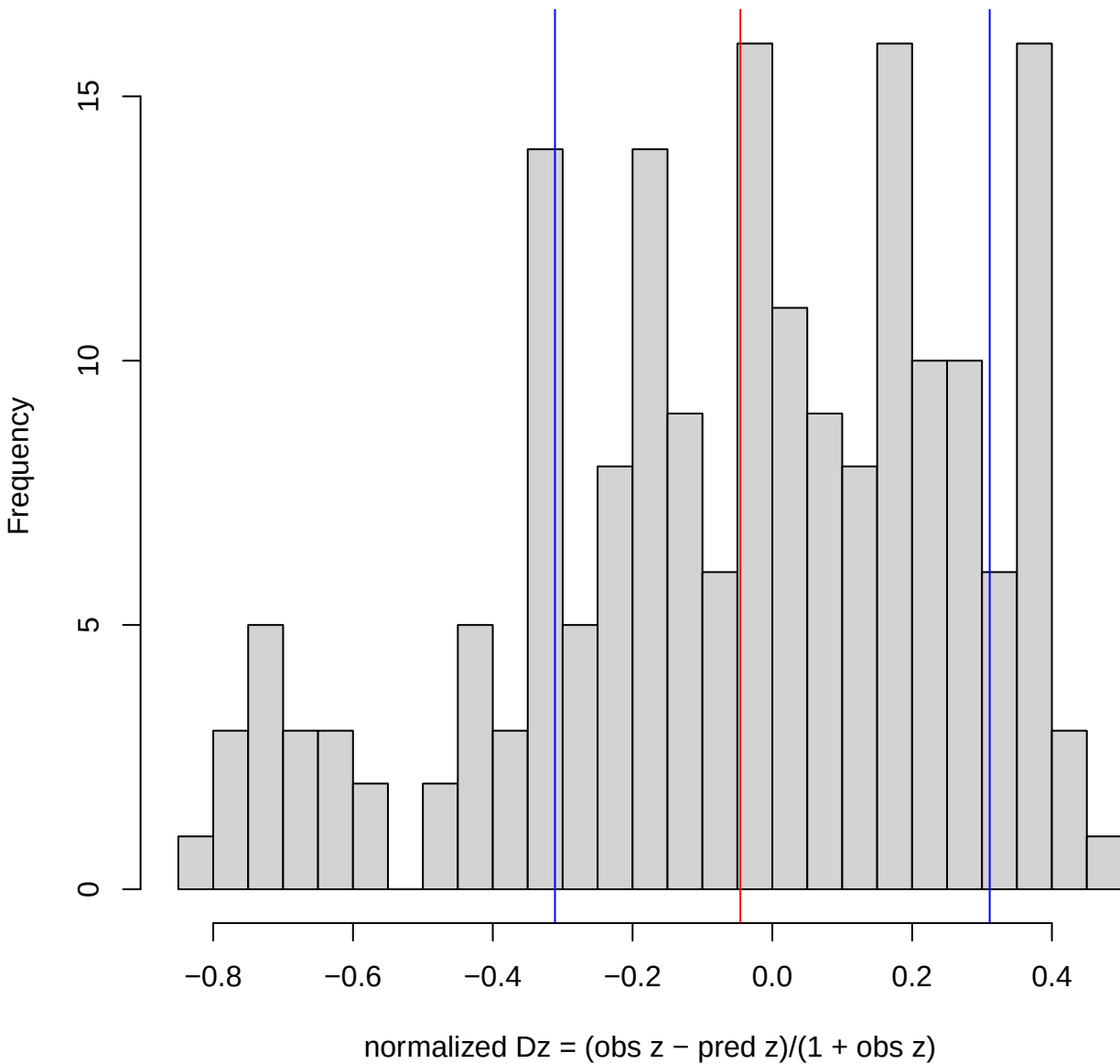
**(blue =  $\pm$ Sigma, red = Bias)**



**Normalized residual distribution (normalized  $D_z = (\text{obs } z - \text{pred } z)/(1 + \text{obs } z)$ )**

**Sigma=0.311, Bias=-0.0458**

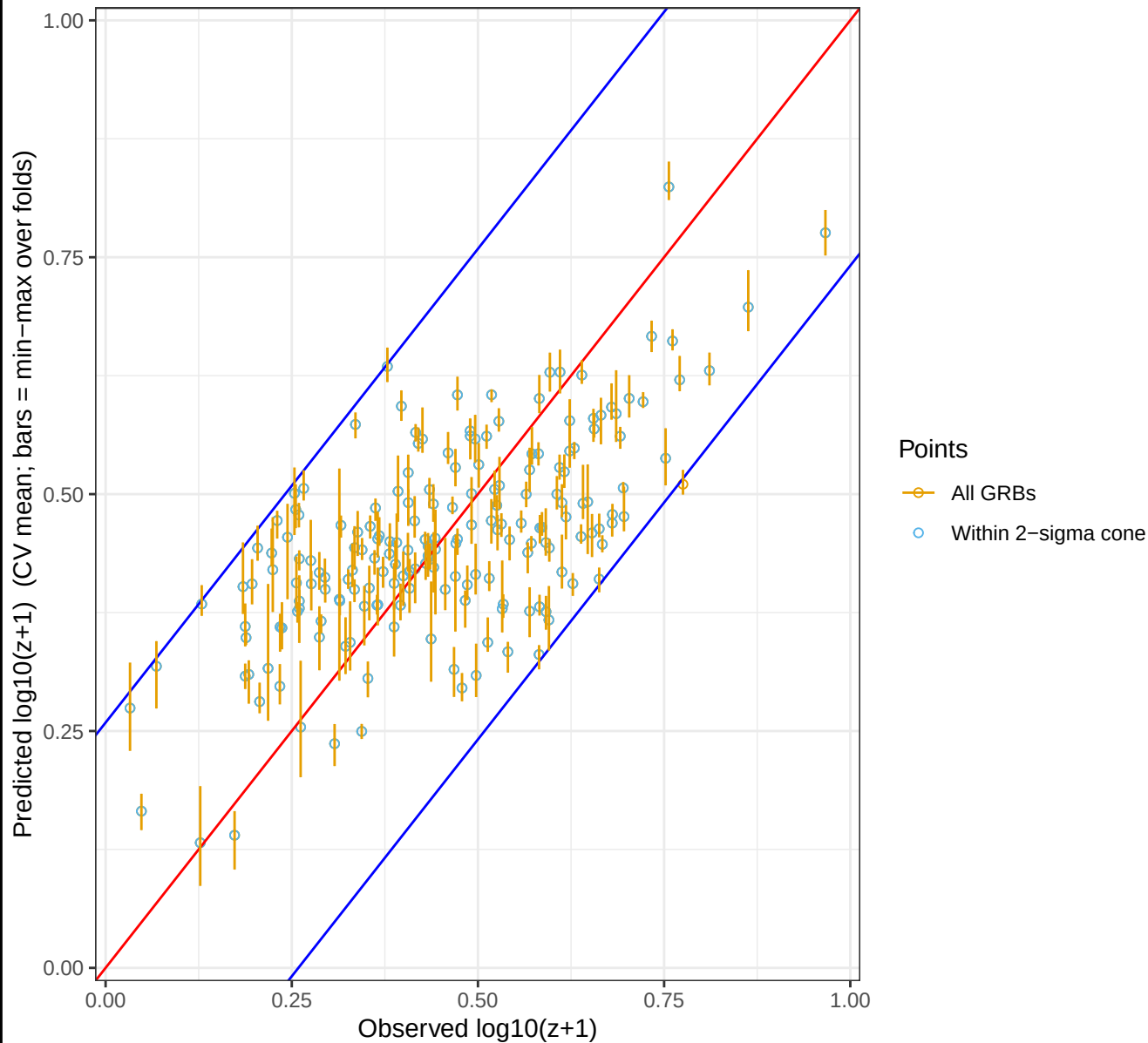
**(blue =  $\pm$ -Sigma, red = Bias)**



Cross-validated predicted vs. observed  $\log_{10}(z+1)$

Sample = 189 GRBs | within 2-sigma cone = 188 (99%)

$r = 0.6499$  |  $\text{Sigma} = 0.129$  |  $\text{RMS} = 0.129$  |  $\text{Bias} = -0.00038$  |  $\text{NMAD} = 0.14$



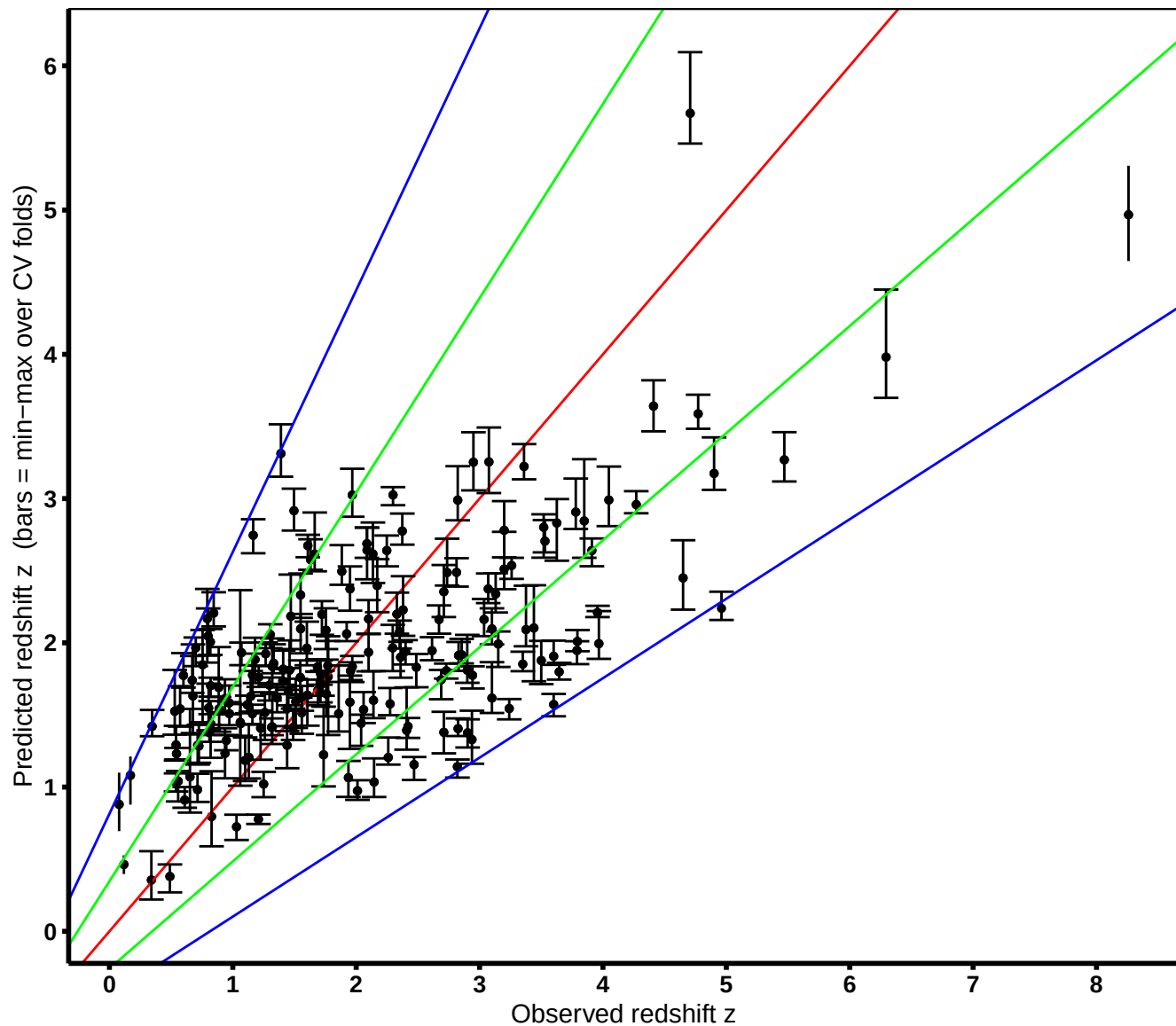


Cross-validated predicted vs. observed redshift  $z$

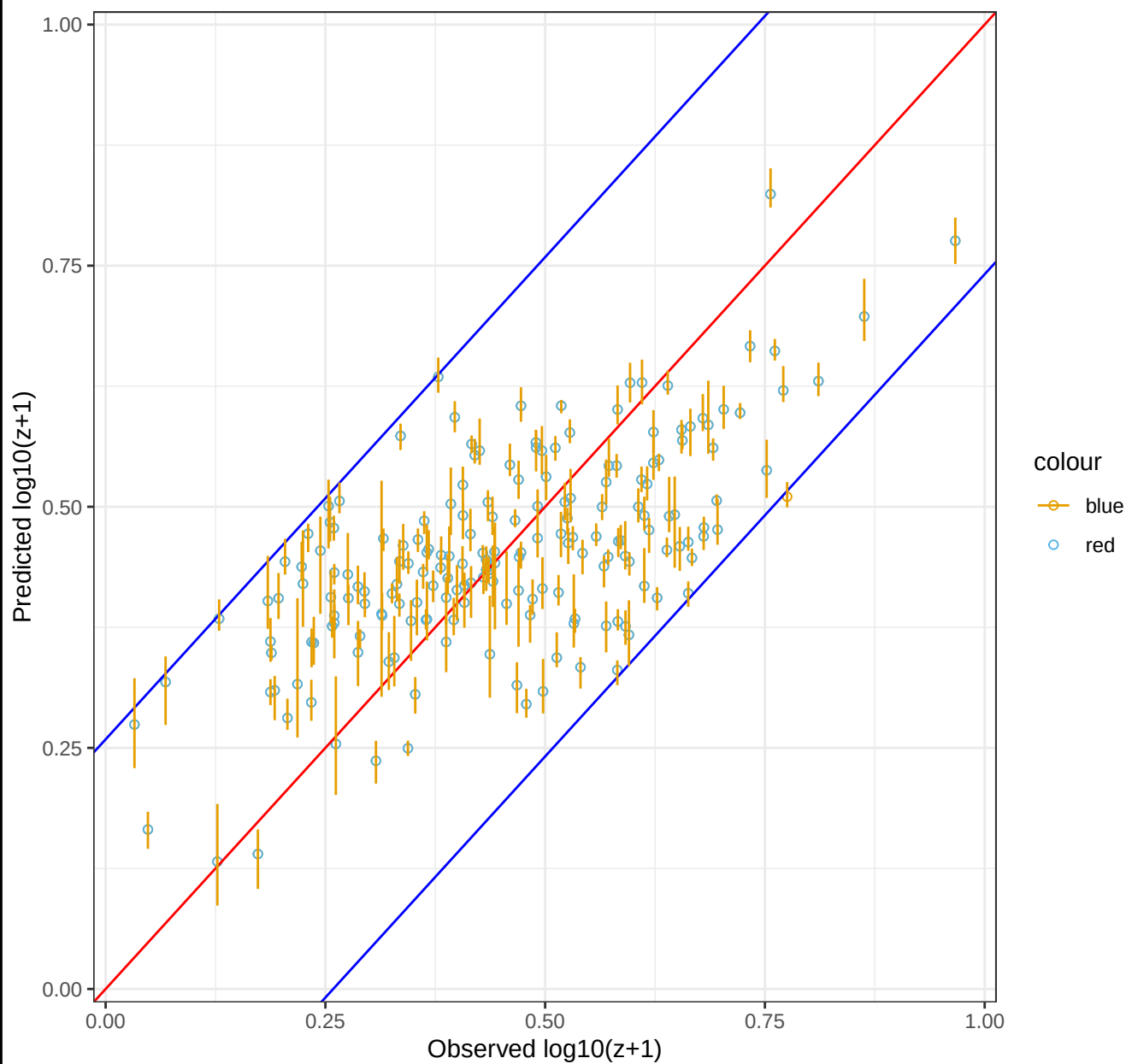
Sample = 189 GRBs | within 2-sigma cone = 188 (99%)

within 1-sigma cone = 126 (67%)

$r = 0.673$  |  $\text{Sigma} = 0.928$  |  $\text{RMS} = 0.94$  |  $\text{Bias} = 0.14$  |  $\text{NMAD} = 1.29$



Predicted vs. observed  $\log_{10}(z+1)$   
(cross-validated)



Predicted vs. observed redshift  $z$   
(cross-validated)

